

Exact Cycle Classification for Every Finite Kac Ring

Route-A structural certificate C170

Abstract

For every ring size and every binary marker configuration, we classify all cycles of the reversible finite Kac ring. The marker product η is the only invariant needed: positive product gives two cycles of length N , while negative product gives one cycle of length $2N$. We derive all fixed counts, the Artin–Mazur zeta, and the finite Koopman determinant.

1 All-marker classification

On $X_N = \mathbb{Z}/N\mathbb{Z} \times \{\pm 1\}$ let

$$T(j, s) = (j + 1, \varepsilon_j s), \quad \eta = \prod_{j=0}^{N-1} \varepsilon_j.$$

One circuit gives $T^N(j, s) = (j, \eta s)$. Since every return time is divisible by N , all states have exact period N when $\eta = +1$ and exact period $2N$ when $\eta = -1$. Hence there are respectively two and one geometric cycles. With $L = N$ or $2N$ and $c = 2N/L$,

$$\#\text{Fix}(T^n) = 2N \mathbf{1}_{L|n}, \quad \zeta_T(z) = (1 - z^L)^{-c}, \quad \det(I - zU_T) = (1 - z^L)^c.$$

The eigenvalues of the counting-measure Koopman permutation are all L -th roots of unity, each with multiplicity c .

2 Boundary

The classification includes $N = 1$. It is an exact finite kinetic toy model, not a target divisor or Hilbert–Pólya construction. Route B is not authorized. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.